EE3032 Introduction to VLSI Design—Final

1. (30%) Answer the following problems.
   (a) List two methods for reducing the power bounce and ground bounce in the IO pads. (4%)
   (b) List major steps of the CMOS full-custom design flow. (6%)
   (c) List two methods for reducing the switching power of a CMOS circuit. (4%)
   (d) Explain the energy consumption of an inverter when the inverter performs a switching. (4%)
   (e) List two major problems of dynamic logic. (4%)
   (f) List two limitations of CMOS domino logic. (4%)
   (g) What are core-limited design and pad-limited design? (4%)
2. (15%) Sketch the boolean function $f = \overline{A(B+C)+DE}$ using each of the following circuit techniques. (1) Domino logic (5%) (2) Fully differential logic (5%) (3) Static CMOS logic (5%)
3. (10%) Explain the operation principle of the circuit shown in Fig. 1.

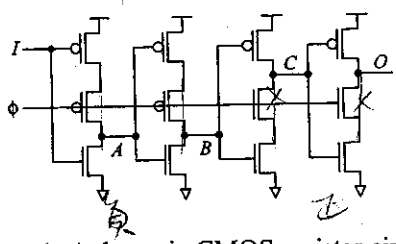


Figure 1: A dynamic CMOS register circuit.

4. (15%) The circuit shown in Fig. 2 results from the implementation of a high-speed one-bit full adder. Complete the following problems.
   (a) What type of CMOS logic style is used to implement this circuit? (5%)
   (b) Suppose that the drain capacitance and gate capacitance of the PMOS and NMOS are 0.2fF and 1fF, respectively. The leakage current of the output nodes $\overline{C}_{n+1}$ and $\overline{S}_n$ is 1nA. Also, the $V_{dd} = 1.8V$ and the switching threshold voltage of the inverter is $1V$. Determine that the lowest speed of the circuit should be operated (10%).
5. (10%) Fig. 3 shows a sequential circuit (pipelined circuit). Assume that your boss asks you to design the circuit such that the circuit can be operated at 100MHz. Also, assume that the clock-to-Q propagation delay, the setup time, and the hold time of the registers R1 and R2 are $1ns$, $1.5ns$, and $1ns$, respectively. How fast of the combinational circuit should be designed?
6. (10%) Figure 4 shows the Schmitt CMOS circuit and its voltage transfer curve. Assume that the Schmitt circuit is designed as $V_{T+} = 0.7V_{dd}$ and $V_{T-} = 0.3V_{dd}$. Also, the size of N_1 is $(W/L)_{n1} = 2\mu m/0.5\mu m$ and the size of P_1 is $(W/L)_{p1} = 3\mu m/0.5\mu m$. What size of the N_3 and P_3 should be designed?
7. (10%) Figure 5 shows the symbolic layouts. Draw the schematic diagrams of the layouts.

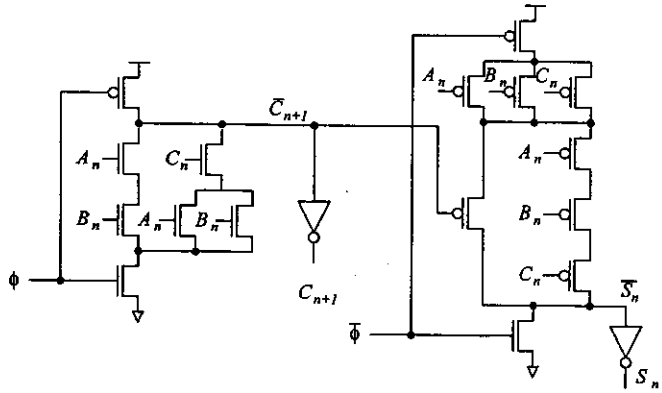


Figure 2: An adder circuit.

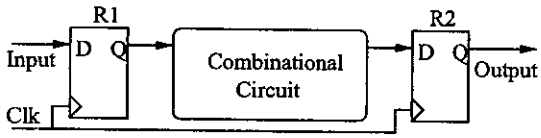


Figure 3: A pipelined circuit.

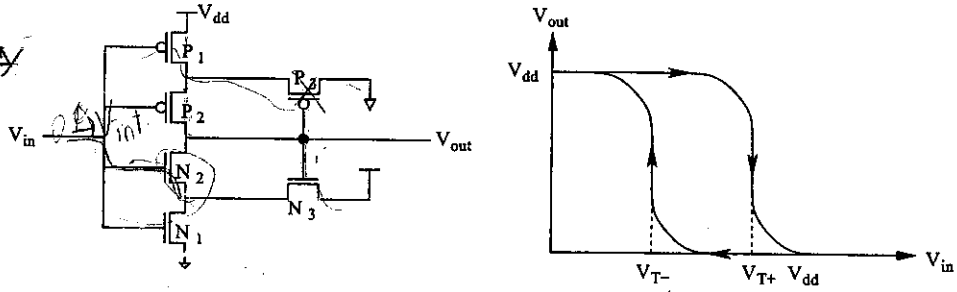


Figure 4: The schmitt circuit.

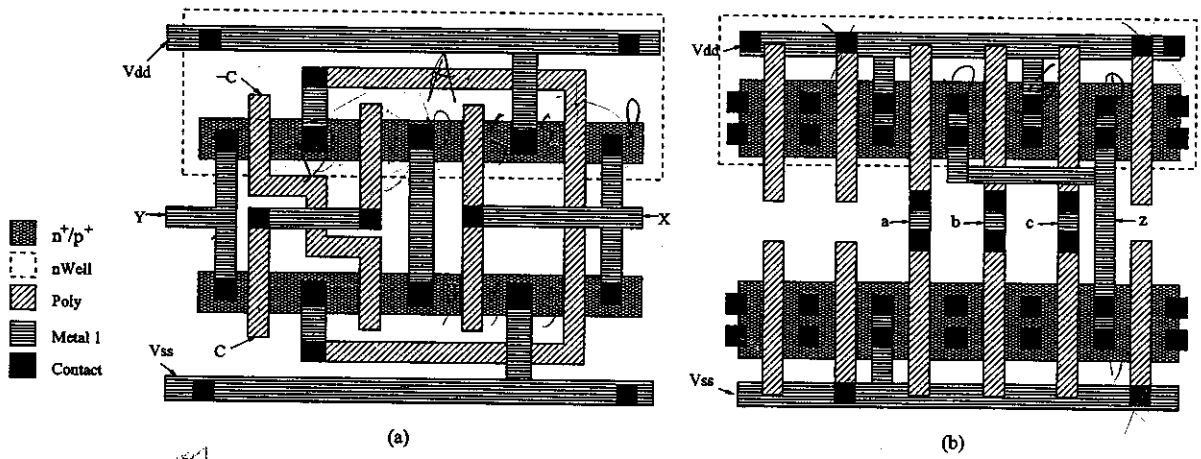


Figure 5: The symbolic layout.

8. (15%) Complete the following problems.

   (a) Figure 6(a) shows a positive-active static latch design, and transistor sizes of the gate in the up-stream of the latch are shown in the figure. Assume that the threshold voltage of the two inverters is $V_{dd}/2$ and the parasitic resistance/capacitance of the transmission gate can be ignored. Also, $\mu_n = 2\mu_p$. Determine the transistor sizes of the inverter 2 such that the latch can work correctly. (10%)

   (b) Use the two latches shown in Figure 6(c) to realize an negative edge-triggered register and draw the corresponding timing diagram. (5%)

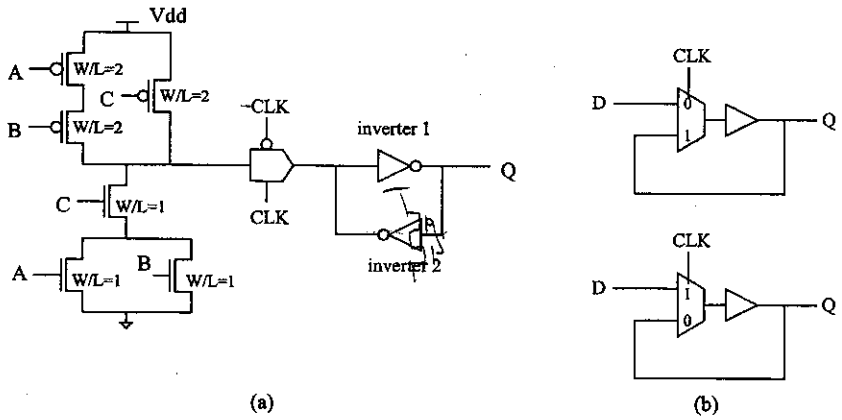


Figure 6: (a)positive-active latch (b) latch clocks the data by edge (c)low-level and low-level latches.